$$\int_0^1 f(x)\,dx = 1$$

$$ak + b(1-k) = 1$$

$$k = \frac{1-b}{a-b}$$

$\because 0 < k < 1$, and $a > b$,

$$\therefore a > 1,\ b < 1$$

(i)

$$\begin{aligned} E(X) &= \int_0^k ax\,dx + \int_k^1 bx\,dx \\ &= \frac{1}{2}ak^2 + \frac{1}{2}b(1-k^2) \\ &= \frac{1}{2}\left(b + \frac{(1-b)^2}{a-b}\right) \\ &= \frac{1-2b+ab}{2(a-b)} \end{aligned}$$

(ii)

$$F(k) = ak = \frac{a(1-b)}{a-b} = \frac{a-ab}{a-b}$$

If $M \in (0, k)$,

$$ak \geq \frac{1}{2}$$

$$\frac{a-ab}{a-b} \geq \frac{1}{2} \implies 2a - 2ab \geq a - b \implies a + b \geq 2ab$$

$$F(M) = \frac{1}{2} \implies M = \frac{1}{2a}$$

If $M \in (k, 1)$,

$$ak \leq \frac{1}{2} \implies a + b \leq 2ab$$

$$ak + b(M-k) = \frac{1}{2}$$

$$bM + 1 - b = \frac{1}{2}$$

$$M = 1 - \frac{1}{2b}$$

$$M = \begin{cases} \dfrac{1}{2a}, & a + b \geq 2ab, \\ 1 - \dfrac{1}{2b}, & a + b \leq 2ab. \end{cases}$$

(iii) When $a + b \geq 2ab$,

$$\begin{aligned} E(X) - M &= \frac{1 - 2b + ab}{2(a - b)} - \frac{1}{2a} \\ &= \frac{b(a-1)^2}{2a(a-b)} \\ &> 0 \end{aligned}$$

When $a + b \leq 2ab$,

$$\begin{aligned} E(X) - M &= \frac{1 - 2b + ab}{2(a - b)} - \frac{2b - 1}{2b} \\ &= \frac{a(b-1)^2}{2b(a-b)} \\ &> 0 \end{aligned}$$

$$\therefore M < E(X)$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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